

```
import pandas as pd
import numpy as np

from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
```

```
data = pd.read_csv("bill_authentication.csv")
```

```
print(data.head())
```

	Variance	Skewness	Curtosis	Entropy	Class
0	3.62160	8.6661	-2.8073	-0.44699	0
1	4.54590	8.1674	-2.4586	-1.46210	0
2	3.86600	-2.6383	1.9242	0.10645	0
3	3.45660	9.5228	-4.0112	-3.59440	0
4	0.32924	-4.4552	4.5718	-0.98880	0

```
x = data.drop('Class' , axis = 1)
```

```
print(x)
```

	Variance	Skewness	Curtosis	Entropy
0	3.62160	8.66610	-2.8073	-0.44699
1	4.54590	8.16740	-2.4586	-1.46210
2	3.86600	-2.63830	1.9242	0.10645
3	3.45660	9.52280	-4.0112	-3.59440
4	0.32924	-4.45520	4.5718	-0.98880
...	...	...	...	...
1367	0.40614	1.34920	-1.4501	-0.55949
1368	-1.38870	-4.87730	6.4774	0.34179
1369	-3.75030	-13.45860	17.5932	-2.77710
1370	-3.56370	-8.38270	12.3930	-1.28230
1371	-2.54190	-0.65804	2.6842	1.19520

```
[1372 rows x 4 columns]
```

```
y = data['Class']
```

```
print(y)
```

0	0
1	0
2	0
3	0
4	0
...	...
1367	1
1368	1
1369	1

```
1370    1
1371    1
Name: Class, Length: 1372, dtype: int64
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.9890909090909091

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	154
1	0.98	0.99	0.99	121
accuracy			0.99	275
macro avg	0.99	0.99	0.99	275
weighted avg	0.99	0.99	0.99	275

Confusion Matrix:

```
[[152  2]
 [ 1 120]]
```

```
#test size=30
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.30)
model = SVC(kernel='linear')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.9927184466019418

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.99	0.99	233
1	0.98	1.00	0.99	179
accuracy			0.99	412
macro avg	0.99	0.99	0.99	412
weighted avg	0.99	0.99	0.99	412

Confusion Matrix:

```
[[230  3]
 [  0 179]]
```

```
#test size=40
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.40)
```

```
model = SVC(kernel='linear')
```

```
model.fit(x_train, y_train)
```

```
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("Classification Report:\n", classification_report(y_test,
y_pred))
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.9890710382513661

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.98	0.99	298
1	0.98	1.00	0.99	251
accuracy			0.99	549
macro avg	0.99	0.99	0.99	549
weighted avg	0.99	0.99	0.99	549

Confusion Matrix:

```
[[293  5]
 [  1 250]]
```